with the important case of its invariant distribution satisfying the partial differential equation

$$\hat{\mathcal{L}}^\dagger \hat{\rho} = 0.$$

Our goal in the following section is to find $\hat{\mathcal{L}}^\dagger$ and solve the PDE for the distribution $\hat{\rho}$ by using our ansatz (7.17). The solution gives us the corrupted invariant distribution sampled by our numerical scheme, allowing us to correct our observed averages to achieve more accurate results. It also gives us a way of evaluating which schemes are best for sampling, by evaluating the error introduced by the discretization.

The terms of the perturbation expansion for $\hat{\mathcal{L}}^\dagger$ can be computed using the Baker-Campbell-Hausdorff (BCH) expansion already introduced in Chap. 3. Recall that for linear operators X and Y, we can write the composition of their exponentials as

$$\exp(hX)\exp(hY) = \exp(hZ),$$

where we have the expansion

$$Z = X + Y + \frac{h}{2}[X, Y] + \frac{h^2}{12}([X, [X, Y]] + [Y, [Y, X]]) + \mathcal{O}(h^3).$$

The notation $[X, Y] \stackrel{\text{def}}{=} XY - YX$ represents the commutator of the operators X and Y. By repeated application of the BCH formula, we can obtain such an expansion for any splitting-based method, however the terms grow in complexity at successive powers of h, requiring commutators of increasing length that make computation cumbersome. Additionally any careful mathematician should question the convergence of this sequence when computing $\hat{\mathcal{L}}$, as we deal only with formal series that offer no guidance on the boundedness of the resulting terms, which will involve high order derivatives in both $\boldsymbol{q}$ and $\boldsymbol{p}$.

To offer some reassurance, let us define the evolution operator for our numerical method as

$$\mathcal{P}_t = e^{t\hat{\mathcal{L}}} = e^{t\mathcal{L}_1} e^{t\mathcal{L}_2} \dots e^{t\mathcal{L}_k},$$

where the numerical method has k-many stages with each $\mathcal{L}_i$ proportional to one of $\mathcal{L}_\mathrm{A}$, $\mathcal{L}_\mathrm{B}$ and $\mathcal{L}_\mathrm{O}$. We can make use of Taylor's theorem to write the identity

$$\mathcal{P}_h = \mathcal{P}_0 + h \left.\frac{\mathrm{d}\mathcal{P}_t}{\mathrm{d}t}\right|_{t=0} + \dots + \frac{h^n}{n!} \left.\frac{\mathrm{d}^n\mathcal{P}_t}{\mathrm{d}t^n}\right|_{t=0} + \frac{h^{n+1}}{n!} \int_0^1 (1-\lambda)^n \left.\frac{\mathrm{d}^{n+1}\mathcal{P}_t}{\mathrm{d}t^{n+1}}\right|_{t=\lambda h} \mathrm{d}\lambda,$$